Plain-language explainer

Ideas behind the coaching questions. EAIF: AI for Finance, Group Assignment AS 2026.

What this document is. Each section explains the idea behind one question on the question sheet, in everyday words, with a small example. You do not need any finance or statistics background. The numbers quoted come from the course data and the starter notebook, and from our first quick look at the files, so treat them as leads that the group will confirm in its own audit.

The big picture in five sentences

- The course gives us **simulated** data about 1,376 companies over 14 years (fiscal years 2012 to 2025). No real firm is behind it.
- Every company publishes a report once a year. The year it covers is its **fiscal year**. The data has one row per company per year, plus one row per company per month for its stock return.
- An **ESG score** (0 to 100) says how responsibly a company behaves. We look at the **Environmental** part.
- We pretend to be an investor who picks stocks on **1 July** each year, holds them for 12 months, and asks: could we have predicted which stocks do well?
- The assignment tests whether we handle the data honestly. The main danger is using information the investor could not have had at the time. This is called **leakage**.

The four tasks of the assignment

- **(a) Audit:** check the data for errors (the course deliberately inserted some) and write a table: the problem, how we found it, how we fixed it, and what it changed in one number.
- **(b) Models:** predict each firm's return over the next 12 months with a simple model (linear regression) and with tree models, and compare them with a naive guess.
- **(c) ESG premium:** ask whether firms with a better Environmental score earned higher returns afterwards, by region and period. A **premium** is extra return earned by one group of firms.
- **(d) Backtest:** replay a strategy on past data as if we had traded it: each July buy the top-ranked 10% of firms (a **portfolio** is the set of stocks held), hold for 12 months and pay trading costs. Compare with buying all firms equally.

A **feature** is an input column the model uses to predict (for example profit margin). The **target** is the number it predicts.

1 Units and rescaling

Belongs to: Q1

A column has a **unit**. Here, Revenue and Market_Cap are in **USD million**. If one value was typed in USD *thousand* instead, it looks 1,000 times bigger than it should be. It is like one runner's time being recorded in seconds while everyone else's is in minutes.

What we saw: 22 firms have revenue above 1,000,000 (USD million), and a few more sit just below that line, so the exact count depends on the cut-off (about 22 to 27). Company_ID 30 has revenue of 1,441,650 in 2012. As USD million that would be USD 1.4 trillion, bigger than any real company. Divided by 1,000 it is 1,441.65, which fits the typical firm (median revenue about 2,300 to 2,400). **Market cap** (the total value of a firm's shares) is about 1,000 times too large for the same firms, so the ratio between market cap and revenue looks normal.

Option on the sheet	What it means	Risk
Divide by 1,000, keep the firms	We treat the values as a unit mix-up and repair them.	If they really are huge firms, we shrink them wrongly.
Keep as is	We treat them as real mega firms.	Size features (such as the logarithm of revenue, a transformation that shrinks big numbers) get extreme outliers: values far from all the others that pull averages.

Option on the sheet	What it means	Risk
Exclude the firms	We remove them completely.	We lose 22 firms (and the audit table should say so).

Why it matters: the audit table (task a) is graded against a list of planted problems. The course deliberately inserted errors into the data. A unit problem is a classic one, and the grader wants to see what we did and what it changed.

2 Near-duplicate rows and why they matter

Belongs to: Q2

A **duplicate** is a row that appears twice. Here the two rows are not exactly equal: Company 1, March 2020 has -27.7587 and -27.7549 (percent). That looks like one measurement stored twice with a tiny rounding difference.

Why it matters: a 12-month return is built by **compounding** the monthly returns, which means multiplying them. Each month is a factor: a return of -27.76% is the factor 0.7224. Take a firm that earned 0% in every other month. Its true yearly return is -27.76%. If the March row is counted twice, the factor is 0.7224 x 0.7224 = 0.5219, which gives **-47.8%**. One extra row makes the loss about 1.7 times larger (-48% instead of -28%).

The two values differ by at most 0.01 percentage points, so keeping the first row or averaging them gives practically the same result. Dropping both deletes the month entirely, which leaves a hole in that firm's year. In the starter, that firm-year would then have only 11 holding months and drop out of the model data (it needs 12), so it is the least attractive.

3 Total return, excess return and what a "target" is

Belongs to: Q3

The **target** is the number the model tries to predict. Here it is a firm's return over the 12 months after the portfolio is formed.

	Definition	Example
Total return	What an investor in the share earned over the period.	Firm earned +8% over the year.
Risk-free rate	What a safe account pays. In the data it is 0.1% to 0.4% per month, which is roughly 1.2% to 5% over a year.	Safe account paid about 1.2% (in a year at 0.1% a month).
Excess return	Total return minus the risk-free rate. It measures the reward for taking risk.	8% - 1.2% = about 6.8%.

The starter notebook predicts the **total** return in task (b). Task (c) asks about the **excess** return. They are close in the training years (about 1 point apart on average) but about 5 points apart in the test years, because the risk-free rate rose after 2021. Every table in the memo should say which one it uses. The handout's decision-time rule names the total return as the target.

What each option changes. With the excess return as the target in (b), the baseline and every error figure shift by the risk-free rate, and (b) and (c) would use the same kind of return. With the total return, (b) follows the starter and the handout wording.

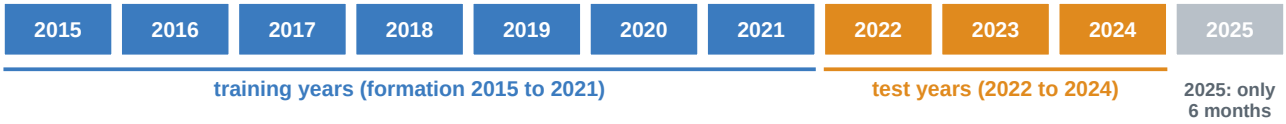
The baseline. The simplest possible forecast is "everyone earns the average of the training years". That average is 1.00%. Its test error (RMSE, the typical size of a forecast error in percentage points) is 36.93. A model is only useful if it beats this number. In the starter, the random forest gets 37.31, so it does *not* beat the baseline. The assignment says that is a legitimate result, as long as we say so. Individual stocks swing a lot, so even the best forecasts are far off. What matters is the difference to the baseline (37.31 against 36.93).

4 Formation year, holding period and training years

Belongs to: Q4

The **formation year** is the year in which the portfolio is formed (on 1 July). The **holding period** runs from that July to the next 30 June. The starter fits the model on formation years 2015 to 2021 (training) and tests it on 2022 to 2024 (test). A random split would be wrong: a model could then learn from 2023 returns to "predict" 2021, which no real investor can do.

Each box is one "formation year": the portfolio is formed on 1 July of that year.



Why 2025 is special. Returns stop in December 2025. A portfolio formed in July 2025 can only be followed for 6 months, so its 12-month return does not exist yet. The starter's model leaves 2025 out (it needs 12 months). Its **backtest** keeps it (6 months), which is why the backtest has 126 months (July 2015 to December 2025) instead of 120 (July 2015 to June 2025). The two options on the sheet are: keep 2025 in the backtest with a note, or stop at 2024 so that every portfolio has a full 12 months.

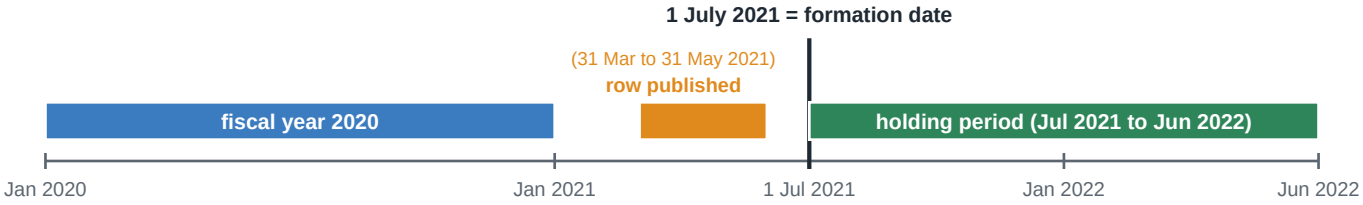
Walk-forward training (task d). In the backtest, the forecast for each formation year may only use *earlier* formation years. For 2018 the model trains on 2015 to 2017. For 2019 it trains on 2015 to 2018, and so on. That mimics a real investor, who never has future data. The first years have very little training data, so we will need our own rule for the minimum number of training years and write it down. The handout leaves that choice to us.

5 Fiscal year, formation year and calendar year

Belongs to: Q5

Three different "years" appear in the data, and mixing them up causes leakage.

- **Fiscal year t:** the year the company reports on. Example: fiscal year 2020.
- **Formation year:** the year in which we may use that report. A fiscal-year-2020 row is published between 31 March and 31 May 2021, so it is used for the portfolio formed on 1 July 2021. Formation year = fiscal year + 1.
- **Calendar year of the return:** the months in which the stock moved. The holding period July 2021 to June 2022 straddles two calendar years, so this option is ambiguous.



The fiscal-year-2020 numbers may be used on 1 July 2021. Nothing published later may be used.

Why the question matters. Task (c) compares "before 2021" and "from 2021". The score for fiscal year 2020 belongs to the formation year 2021. If the split is defined on the *fiscal* year, that score falls in the "before" group. If it is defined on the *formation* year, it falls in the "from 2021" group. The assignment says the data contains one genuine relationship, which should show up in one cell of the region-by-period table, so moving a year of data between the two periods can hide or dilute it. Example: the score for fiscal year 2020 (formation year 2021) is in the "from 2021" group under option 1, but in the "before" group under option 2.

6 Standardising the score (and the task (c) method)

Belongs to: Q6

Standardising (z-score). Subtract the average and divide by the spread: z = (score - average) / standard deviation. After this, the average is 0 and one unit means "one standard deviation higher". In the data the within-year spread of the Environmental score is about 17 points, so z = +1 means about 17 points above the year's average. We standardise within each year so that a year with generally higher or lower scores does not distort the comparison.

The option choice. "Within year, across all firms" compares a firm with everybody. "Within year and region" compares it only with its regional peers. Because task (c) reports results by region, standardising within region would change what "one unit" means in each region.

Example of the option choice: if the spread in Europe is 10 points and in Asia 25, one unit would mean 10 points in Europe but 25 points in Asia, so the units would differ between regions.

Only needed later, not for the question: the method behind task (c)

Regression with industry-by-year controls. A regression fits a line through a cloud of points. Its slope (the **coefficient**) says how much the return changes when the score goes up by one unit, holding other things fixed. "Industry-by-year controls" means we compare each firm only to firms in the same industry in the same year (for example Energy firms in 2019). This stops us from mistaking an industry boom for an ESG effect.

Standard errors. A coefficient is only an estimate. The standard error says how uncertain it is. If the estimate is less than about two standard errors from zero, we cannot tell it apart from noise. Two ways to get honest standard errors: **clustering by company** (we tell the software that the rows of one company belong together, because one company's years resemble each other and do not count as separate evidence; this is unrelated to clustering in machine learning), or the **bootstrap** (draw rows at random from our own data, with replacement, thousands of times, and see how much the answer jumps).

7 Leakage and the Analyst_Outlook column

Belongs to: Q7

Leakage means the model sees information it would not have in real life. It is like a student who studies with the answer sheet: great marks, no real skill. The decision-time rule is the test: *was this value in a file that existed on 1 July, with the value it had then?*

Our finding. Analyst_Outlook is a score from 1 to 5. We measured its rank correlation with the return in several 12-month windows. A **rank correlation** (Spearman) asks whether firms with a higher rank on one variable also tend to have a higher rank on the other. It runs from -1 to +1, and 0 means no link.

Here t is the fiscal year of the row. Example for the portfolio formed on 1 July 2021 (the fiscal year 2020 row):

12-month window compared with Analyst_Outlook	Rank correlation
The 12 months after formation, Jul 2021 to Jun 2022 (the thing we try to predict)	**0.67**
The 12 months before formation, Jul 2020 to Jun 2021	-0.01
The 12 months before that, Jul 2019 to Jun 2020	0.01

Real analyst ratings predict returns only weakly. A column that agrees so strongly with the future and with no other window was probably stamped with future information. A model that used it would look great on paper and be useless in practice.

Option on the sheet	What it means
Yes, that is enough	We report it in the audit table (how found, that we excluded it) and list it as "excluded" in the feature table. The feature table shows every candidate feature and the date from which it is known.
Also show a model with it included	A warning demonstration: it shows how misleadingly good a leaking model looks. The options add to each other: tick all that apply.

8 The "effect on one headline number"

Belongs to: Q8

A **headline number** is one figure that summarises a result, for example the average 12-month return, the model's test error (RMSE), or the Sharpe ratio. For each problem in the audit table, we show what the number was *before* and *after* the fix. Example row: "Near-duplicate returns; found by counting repeated firm-months; fixed by keeping one row; the average 12-month return changed from X to Y."

The **Sharpe ratio** is the average excess return divided by its ups and downs (volatility), scaled to a year. It measures return per unit of risk. Higher is better.

9 Page and slide limits

Belongs to: Q9

This is a rule question, no concept needed. The handout says "3-page memo (plus appendix of any length)" and "at most 8 slides, 10-minute talk". It does not say whether tables and figures count towards the 3 pages, or whether the title slide counts as one of the 8. Example: if everything counts, the audit table and the results tables compete with the text for three pages, and most detail has to go to the appendix. If only text counts, they can stay in the memo body.

10 Libraries, Colab and run time

Belongs to: Q10

Colab is Google's free online notebook: our code runs on Google's computers. A **library** is a toolbox of ready-made code; **pip** is the command that installs one. **scikit-learn** has the models of task (b) and is already used in the starter. **statsmodels** fits regressions and prints standard errors, which task (c) needs (without it we would have to code them by hand). **lightgbm** and **xgboost** are fast versions of gradient boosting (see below). The handout requires that the notebook runs top to bottom in Colab without errors, which is why we ask about libraries and about a time limit: tuning models again for every year means many model fits, and that takes time.

Option on the sheet	What it means
Plan is fine	We may use scikit-learn, statsmodels and possibly lightgbm.
Only libraries already installed in Colab	We must check each import before using it.
Only scikit-learn and the starter's imports	We would code standard errors for task (c) ourselves.

Optional background for tasks (b) to (d)

Not needed for the question, but useful for the project.

Name	In plain words
Linear regression	Finds one weight per input and adds them up (like a recipe). Easy to read, can only draw straight-line relationships. The handout asks for standardised inputs, which makes the weights comparable.
Random forest	Builds hundreds of decision trees (flowcharts of yes/no questions such as "is profit margin above 10%?") and averages their answers. Each tree sees a random part of the data.
Gradient boosting	Also trees, but built one after another. Each new tree tries to fix the mistakes of the previous ones.
Overfitting	The model memorises the training data (including its noise) and does badly on new data.

Tuning and cross-validation. A model has **settings** chosen by us, such as how deep the trees may grow (hyperparameters). We choose them with **cross-validation**: cut the training data into 5 parts, train on 4, check on the fifth, and rotate. **GroupKFold** keeps all rows of one company in the same part, so the model cannot recognise a company it saw in training. Inside the training years the parts mix different years, so the cross-validation score is a little optimistic. That is why the test years come last and are used once.

Starter result (random forest)	RMSE (percentage points)
Train (in sample)	28.78
Cross-validation (training years)	29.03
Test (2022 to 2024), used once	37.31

Starter result (random forest)	RMSE (percentage points)
Naive baseline on the test years	36.93

Cross-validation looks better (29.0) than the test (37.3). The starter's own text blames a market downturn in the test years, but the data shows the opposite: the average 12-month return was **+21.6%** in the test formation years 2022 to 2024 against **+1.0%** in the training years. The training-mean baseline (1.00%) is therefore far too low for the test years, and that is what makes the test errors large and the R squared negative. Task (b) should say this plainly, and not only report the numbers.

Tune once vs re-tune each year (task d). The handout accepts both. Re-tuning every year is more realistic but slower. Tuning once is quicker, but it uses information from later years when choosing the settings for early forecasts, so the write-up must state this.

11 Use of AI tools

Belongs to: Q11

Rules differ between courses, and the handout transcription we have does not mention AI tools. The handout says that results.json lets grading check the numbers against the notebook and the memo before a person reads either, and Prof. Matter's feedback on the earlier take-home says explanations are also read by a person. The handout also states that every group member must be able to explain every cell of the notebook. It is best to know the rule before we start, and to keep a short log of where AI helped.

12 Notebook version

Belongs to: Q12

Our copy of the starter notebook was saved in September and still has a placeholder where the deadline should be. The handout PDF now gives the dates (27 November for the memo, notebook and peer evaluation; 7 December for slides; 8 December for presentations). Asking avoids building on an outdated copy.

A Quick glossary

Belongs to: terms used in several sections

Term	Meaning
Fiscal year	The year a company's yearly report covers.
Leakage	A model uses information that would not have been available at decision time.
Backtest	Replaying a strategy on past data as if it had been traded.
Baseline	A deliberately simple forecast (here: the training average) that every model must beat.
Compounding	Multiplying monthly return factors to get a yearly return.
Volatility	How much returns go up and down; technically the standard deviation of returns, usually per year.
Point-in-time	Using only what was known on the decision date, with the value it had then.
Formation date	1 July each year: portfolios are picked and held until the next 30 June.
Survivorship bias	Testing only on firms that still exist, which hides the ones that failed. The file universe_current.csv has only survivors (987 of 1,376 firms).
Restated scores	ESG scores as a download today shows them, revised with hindsight, so they are not what an investor knew back then.
RMSE	Typical size of a forecast error, in the units of the target (percentage points of return). Lower is better.
R squared	Share of the variation the model explains. 0 is as good as the average; negative is worse than the average.
Bootstrap	Draw rows at random from our own data, with replacement, thousands of times, to see how much a result moves by chance.
Sharpe ratio	Average excess return divided by volatility, scaled to a year: return per unit of risk.
Turnover	Share of the portfolio replaced at each rebalancing. Each unit costs 20 basis points (0.20%).